

Question 1.7.1 When doing Knn classification we split our data into training and test sets.

Why do we divide our data into training and test sets? Or in other words what is the point of the training set? What is the point of the test set? **(7 points)**

Hint: Check out this [section](#) in the textbook.

Note: Please answer both questions

The training set is used to train the model and help the model learn the patterns and relationships within the data. The test set evaluates the model's performance and how well it generalizes to new data.

Question 1.7.2 Why do we only want to use the test set once? **(3 points)**

We use the test set once in KNN classification to avoid overfitting and getting an unbiased estimate of the model's performance on new data. Using the test set multiple times can hinder the model from generalizing well to other data.

Question 1.8. Why do we choose k to be an odd number in k -NN? Explain. **(10 points)**

If k is an even number, there is a chance that each of the two classes gets the same amount of votes, making it difficult to classify any new data points. When k is odd, there is always a majority class among the k nearest neighbors, ensuring a clear way to classify the data points.

